TASK–1 REPORT

Cognifyz Machine Learning Internship

Task Title	Restaurant Rating Prediction
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Domain	Machine Learning

1. Objective

The primary goal of this task is to design a machine learning model that can accurately predict the aggregate rating of a restaurant based on multiple influencing features.

2. Dataset Description

The dataset includes comprehensive restaurant-related information such as city, cuisines, price range, votes, delivery options, and aggregate ratings.

3. Methodology

Data preprocessing involved handling missing values and encoding categorical features. Exploratory analysis was performed to understand relationships among variables.

4. Model Development

A Linear Regression model was trained using an 80:20 train-test split to learn the relationship between input features and restaurant ratings.

5. Model Evaluation

The model was evaluated using Mean Squared Error (MSE) and R-squared metrics, which indicated acceptable predictive performance.

6. Results & Insights

Features such as online delivery availability, price range, and customer votes significantly influenced restaurant ratings.

7. Conclusion

The task objectives were successfully met. The developed model provides a reliable baseline for restaurant rating prediction.

8. Future Scope

Future improvements may include using ensemble models, hyperparameter tuning, and deploying the solution as a web application.

9. References

• Scikit-learn Documentation
• Pandas Documentation
• Cognifyz Internship Learning Resources